

Figure 9.11: Cryptococcus **(a)** capsule staining **(b)** On SDA **(c)** On BSA

Clinical Features

The clinical features of Cryptococcosis depend upon the anatomical sites.

i. Pulmonary Cryptococcosis

The respiratory route is usually the portal of entry for propagules in Pulmonary Cryptococcosis that subsequently disseminate to extra pulmonary sites. The symptoms are dry cough, dull chest pain and milder or no fever with small gelatinous granules all over the lungs.

ii. CNS Cryptococcosis

This is an infection of brain and meninges leading to Meningoencephalitis. Nitrogenous source such as asparagines and creatinine present in cerebrospinal fluid enrich the yeast. The symptoms are nausea, dizziness, impaired memory, blurred vision and photophobia. The enlarged granulomatous cerebral lesions are called **cryptococcoma**.

iii. Visceral Cryptococcosis

This infection usually spreads from a primary focus to invade the optic nerve and meninges. Visual loss in patients is due to intra cranial pressure. There are two distinct patterns of visual loss namely; rapid visual loss (within 12 hrs) and slow visual loss (within weeks to months).

Laboratory Diagnosis

i. Samples

Specimens collected are mainly serum, CSF and other body fluids.

a. Direct Examination

10% Nigrosin or India ink staining, Gram staining and LPCB are used to visualize the yeast cell.

Biopsy material is stained with periodic acid - Schiff and Gomoris's methylamine silver stain to observe the fungal cells in the tissue. Round budding yeast cells with a distinct halo gelatinous capsule can be seen (Figure 9.11a). Gram positive budding yeast cells are demonstrated by Gram staining.

b. Fungal Culture

The clinical specimens can be cultured on Sabouraud dextrose agar, Bird Seed agar and incubated at 37°C. The colonies are mucoid, cream to buff - colored in SDA (Figure 9.11b), whereas brown colored due to conversion of the substrate into melanin by Phenoloxidase in BSA (Figure 9.11c).

ii. Treatment

1. Amphotericin B, Flucytosine is given together as induction and maintenance therapy.
2. Fluconazole is also recommended.

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